

Salvatore Gutierrez

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EDUCATION

California State University, Bakersfield

Bakersfield, CA

Bachelor of Science, Computer Science with Minor in Math - Statistics

August 2019 – December 2024

Completed Coursework: Computer Architecture, Computer Architecture II, Algorithms, Artificial Intelligence, Advance Artificial Intelligence, Mobile/Web Development, Programming Languages, Probability Theory, Discrete Structures, Database Sytems, Algorithm Analysis, Operating Systems, Computer Networks, Applied Statistical Computing, Multivariate Analysis 1-2, Calculus 1-3.

SKILLS

Programming Languages/Frameworks: C++, JavaScript, Python, SQL, R, SAS, React, ExpressJs, TypeScript.

PROJECTS

Social Media App – Full-Stack Web App

Languages and Tools – TypeScript, React, NodeJS, AWS, Postgres, Socket.io, MongoDB, Redis, Oauth, Docker, Linux

- Developed a full-stack social web application where users can create posts, add friends, and connect with strangers in real time based on tags. Built the frontend with TypeScript, React, and Tailwind CSS for a responsive and visually appealing UI.
- Implemented a scalable backend using Google OAuth for authentication, Redis for session management, PostgreSQL for storing user data, and AWS S3 & CloudFront for efficient media storage and distribution.
- Designed a real-time private chat system with WebSockets and Socket.IO, leveraging Redis queues and Pub/Sub to efficiently manage WebSocket connections across multiple servers for improved scalability.

Flashcard app – Mobile App

Languages and Tools Used: JavaScript, CSS, React native, NodeJS, MongoDB, C++, OpenCV, cpp-httpilib, BOOST,linux

- Design and built an Object-Oriented API to detect text in an image using C++, OpenCV, and cpp-httpilib. Used modern c++ techniques like smart pointers, lambdas, BOOST library.
- Users can scan text from their phone camera, which will generate flashcards with my API. Flashcards sets and quizzes can be created based off their flashcards, in order to test their knowledge.
- Utilized the MERN tech-stack, with React-Native for the front-end, and NodeJS and ExpressJS to create an API to store user data in MongoDB with AWS as a cloud provider.

DinoDash – c++ multiplayer game

Languages and Tools Used: C++, X11, OpenGL, Linux, SFML, Google Cloud, Docker, Kubernetes, Github, CI/CD

- C++ multiplayer game, used OpenGL and x11 for rendering graphics and a window system.
- Design and implemented my own classes to simulate 2d frameworks for rendering graphics and inputs.
- Developed a distributed system for client and server communication. A matchmaking server to handle players queues using TCP/IP sockets, and a game server using UDP sockets for real time updates in games.
- Containerize both servers using Docker, scaled servers horizontally using Kubernetes. Deployed the matchmaking server with dynamic scaling based on queue size, used a load balancer to handle traffic. Game-server was scaled dynamically, a job was deployed using Kubernetes API based on players in queue.

ACADEMIC RESEARCH/WORK EXPERIENCE

Hybrid GARCH-LSTM-based recurrent neural network for short-term flu forecasting

Languages and Tools Used: Python, Pytorch, TensorFlow, matplotlib, pandas, scikit-learn, NumPy, Linux

- Cleaned and process the dataset to handle outliers and missing values. Applied feature engineering using forward and backward selection to find optimal predictors. Transformed the time-series dataset into a supervised learning dataset with lagged features to feed into an LSTM, MLP, RF, and OLS.
- Implemented a bidirectional LSTM model inspired by prior research using both PyTorch and TensorFlow, achieving results consistent with the original study.

Electrocardiogram Simulator, Kern County Hospital

Languages and Tools Used: Python, Tkinter, numpy, scikit learn, github, scipy, matplotlib, Linux, Docker

- Design and developed an object-oriented electrocardiogram simulator for the Kern County Hospital.
- Implemented waveforms using mathematical models derived from ECG research papers, validated against patient data with input from medical professionals.

- Developed a GUI in Python with Tkinter that allows doctors to simulate various cardiac conditions, providing a training platform for medical students. Used Matplotlib for wave visualization and animation, enabling real-time ECG data generation and interactive simulations.